

Changsheng Sun

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PROFILE

AI researcher working on **coding agents**, **AI safety**, and **trustworthy machine learning**, with hands-on work across CNNs, GNNs, 3D point-cloud models, LLMs, and LLM agents. Recently led a long-horizon agent benchmarking project; prior work covers robustness, risk-aware optimization, explainability, and LLM jailbreaks.

EDUCATION

Ph.D. in Computer Science (Advisor: Prof. Jin Song Dong)	National University of Singapore, 2026
M.Comp. in Artificial Intelligence	National University of Singapore, 2021
B.Eng. in Computer Science	Xidian University, China, 2019

EXPERIENCE

SoTA AI Lab, Zhihu Inc. **Singapore**
Research Intern; Project Lead & First Author, SWE-Continuum Jun. 2026 – Sep. 2026

- Led a 10-person team, incl. researchers from Moonshot AI (Kimi) and CityU HK, on a coding-agent benchmark.
- Designed and built from scratch a new task paradigm, task curation pipeline, and metric suite for real-world long-horizon agent evaluation, with a verifiable DAG task compiler for realistic multi-step trajectories.
- Profiled agent trajectories beyond pass/fail: capability acquisition, preservation, regression/recovery, and cost.

National University of Singapore **Singapore**
Ph.D. Researcher, Machine Learning & Trustworthy AI Jan. 2022 – Present

- **GNNs**: developed risk- and direction-aware GNN explanations; co-developed DiGCN (NeurIPS), now in PyG.
- **3D vision**: co-developed PointCVaR, risk-optimized outlier removal for robust point clouds (AAAI Oral).
- **LLM safety**: corresponding author of a survey on modern LLM jailbreak attacks and defenses (ACL Findings).

Research Assistant Jun. 2021 – Dec. 2021

- **CNNs**: built InputReflector to repair distribution-shifted image inputs without retraining (+30.78% accuracy, ASE).
- **Data**: built Spark/ClickHouse pipelines on TB-scale parking/telecom data for spatiotemporal forecasting.

Singtel-NUS Cyber Security R&D Lab **Singapore**
Research Intern Jan. 2020 – Jun. 2020

- Built SelfChecker, a deployment-time CNN monitor catching 60.56% of wrong predictions at a 2.04% false-alarm rate and 34.98 ms overhead, incl. autonomous-driving models (ICSE).

JD Intelligent Cities Research, JD.com, Inc. **Beijing, China**
Research Intern Jan. 2018 – Jun. 2018

- Prototyped a Ray-based distributed DQN system for large-scale spatiotemporal data, advised by Prof. Yu Zheng.

SELECTED PUBLICATIONS

[Google Scholar](#): 700 citations, h-index 7 [†] First author * Corresponding author

[1] SWE-Continuum: Multi-Stage Profiling of Behavioral Continuity in Long-Horizon Agents[†]. *Under review at ICLR, 2027.*

[2] From Attack Surfaces to Actual Operations: A Survey of Modern LLM Jailbreaks*. *ACL Findings, 2026.*

[3] Risk-Aware Robust Graph Network Explanation[†]. *Under review, 2026.*

[4] Ignoring Directionality Leads to Compromised Graph Neural Network Explanations[†]. *IEEE S&P Workshops, 2025.*

[5] PointCVaR: Risk-optimized Outlier Removal for Robust 3D Point Cloud Classification. *AAAI (Oral), 2024.*

[6] Repairing Failure-inducing Inputs with Input Reflection. *ASE, 2022.*

[7] PIANO: Influence Maximization Meets Deep Reinforcement Learning. *IEEE TCSS, 2022.*

[8] Self-Checking Deep Neural Networks in Deployment. *ICSE, 2021.*

[9] Digraph Inception Convolutional Networks. *NeurIPS, 2020.*

TECHNICAL SKILLS

Models: CNNs, GNNs, 3D point-cloud networks, deep RL (DQN), LLMs, LLM coding agents.

Research Areas: Agent evaluation, AI safety and robustness, risk-aware (CVaR) optimization, explainability.

Engineering: Python, SQL, PyTorch, TensorFlow, PyG/DGL, Docker, Harbor, Ray, Spark, ClickHouse.

SERVICE & AWARDS

Reviewer / PC: NeurIPS, ICML, ICLR, AAAI, WWW, FSE, ASE, ISACE (2022–2027).

Teaching: TA for Formal Methods (CS5232/CS4211) and Software Testing (CS4218), 2022–2024.

Awards: NUS Research Scholarship; SoC Tuition Fee Support Award; IEEE S&P, AAAI, and ASE travel grants.